ABN SEAL COLLEGE

COOCHBEHAR

PANCHANAN BARMA UNIVERSITY

Adaptations of Hydrophyte, Halophyte and Xerophyte

B.Sc. 2nd Semester

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Hydrophytes :

Ø Definition :

Hydrophytes are plants that grow completely or partially in water or in very wet habitats. These plants are specially adapted to survive in aquatic environments.
examples- Lotus, Hydrilla, water Hyacinth.

Ø Types :

Types of Hydrophytes

① Floating Hydrophytes
- Free floating
- Rooted floating

② Submerged floating Hydrophytes

③ Rooted submerged Hydrophytes

④ Rooted emergent Hydrophytes or Amphibious Hydrophytes

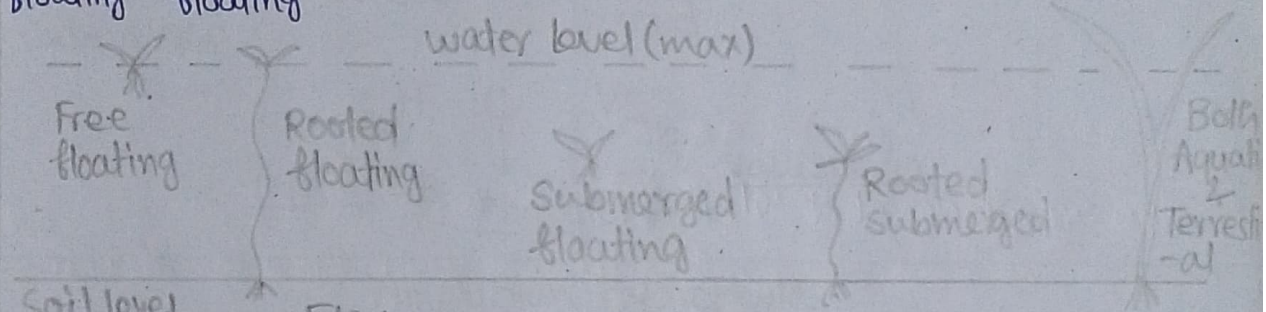


Fig: —

Free floating hydrophytes

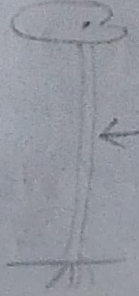
- Float freely on the water surface
- not attached to the soil
- Roots are poorly developed
- Plant body remains buoyant.

P → Pistia
W → water hyacinth

Rooted floating Hydrophytes

- These plants remain rooted in mud.
- But leaves float on the water surface
- Long petioles help leaves float
- Stomata present on upper surface of leaf.

N → Nymphae
L → Lotus



Submerged floating Hydrophytes

- Plant remains completely submerged.
- not attached to soil
- Thin & flexible body
- Stomata generally absent.

C → Ceratophyllum spp.
U → Utricularia

Rooted submerged hydrophytes

- Plant remains completely submerged
- root attached to soil
- roots absorb minerals from mud.
- leaves are narrow and ribbne like.

H → Hydrilla spp
C → Chara spp
V → Vallisneria

Amphibian Hydrophytes

- These plants can live both in water and on land.
- show characteristics of both aquatic & terrestial plants.

T → Typha
N → Neptunia

✿ Ecological adaptations of Hydrophytes

Hydrophytes are the plants that grow in water or in places with excess moisture. Since they live in aquatic environments, they develop several structural and physiological adaptations that help them survive and complete their life cycle in water. These adaptation helps in ~~bou~~ buoyancy, gaseous exchange, absorption in minerals and protection from water currents.

① Root Adaptation –

In moist hydrophytes, the root system is poorly developed because water and dissolved minerals are directly available in the sorrounding environment. Roots mainly function as anchorage rather than absorption.

Root hairs are usually absent or very few in number. In free floating hydrophytes like pistia and eichhornia, roots are highly reduced and hang freely in water.

② Stem adaptations:

1) Stems of hydrophytes are soft, flexible & spongy.
2) mechanical tissues are poorly developed because water itself supports the plant body.
3) Stem contains large air-filled spaces known aerenchyma This air chamber help in buoyancy and fascilate the exchange of gases.
4) Flexibility of the stem helps the plant withstand water currents and waves without damage.

③ Leaf adaptations:

1) leaves of floating hydrophytes are broad, flat, and covered with a waxy coating, this waxy coating prevents wetting and decay.
2) In submerged hydrophytes, leaves are thin, narrow, or finely dissected to reduce resistance against water currents.
3) Petioles are long and flexible in rooted floating plants such as lotus and water lily, helping the leaves remain on the water surface.

④ Stomatal Adaptations:

The distribution of stomata differs according to the habitat of the plant.

① In floating hydrophytes, stomata are present only on the upper surface of the ~~plant~~ leaves because the lower surface remains in contact with water.

② In submerged hydrophytes, stomata are generally absent since ~~gas~~ gaseous exchange occurs directly through the plant surface.

⑤ Cuticle Adaptations:

① Hydrophytes usually possess a very thin cuticle or completely lack it because there is no danger of excessive water loss. Thin cuticle helps easy absorption and exchange of water and gases through the plant surface.

⑥ Vascular Tissue Adaptations:

① Vascular tissues, especially xylem, are poorly developed in hydrophytes because water is abundantly available around the plant body. The requirement for conduction of water is therefore very low.

⑦ Mechanical Tissue Adaptations:

① ~~Mechic~~ Mechanical tissues like sclerenchyma are greatly reduced because aquatic plants donot need much support to the plant body.

⑧ Physiological Adaptations:

① Hydrophytes can tolerate low oxygen conditions present in waterlogged habitats.

② presence of Arenchyma helps in storage and transport of oxygen to submerged parts of the plant.

Halophytes:

Halophytes are the plants that grow on saline soils, marshes, mangrove swamps, and coastal regions where the salt concentration is very high. Since they live in high saline environments, they develops several structural and physiological adaptations, that help them to survive and complete their life cycle in that environm-ent.

① Root adaptations:

① Halophytes possess well developed root systems for proper anchorage in loose muddy soil and for efficient water absorption.

② In mangrove plants, several respiratory roots called pneumatophores develop. These roots grow upword above the soil surface & help in gaseous exchange in waterlogg and oxygen defficient soils.

③ Pneumatophores contain small pores called ~~lenticell~~ lenticels through which oxygen enters.

② Stem adaptations:

① Stem of many halophytes are thick, succulent.

② Succulence helps in storing water and diluting the salt concentration inside the plant tissues.

③ Leaf adaptations:

① Leafs of halophytes are usually thick, leathery and succulent.

② The thick leaves store water and help to reduce the harmful effects of salt concentration.

③ Many halophyte possess reduced leaf area to minimize transpirations.

④ Salt secreting structure:

① Many halophytes possess special salt glands or salt hairs on leaves and stems.

② These structure remove excess salt absorption from the soil, salt crystals are often seen on leaf

Sunken ✓ Sunkane ✗

surface after secretion.

⑤ cuticle adaptations:

① Halophytes usually possess a thick cuticle covered with wax. This reduces transpiration and prevents excessive water loss under saline conditions where water absorption is difficult.

⑥ Stomatal adaptation:

① Stomata are few in number and often sunken in halophytes. Suncane stomata reduce the rate of transpiration and help conserve water.

⑦ Physiological adaptation:

① Halophytes maintain high osmotic ~~pressure~~ concentration in their cell sap by accumulating salts and organic substances. This helps them absorb water from saline soils.

② Many halophytes can tolerate and store large quantities of salt within vacuoles without damaging the protoplasm.

⑧ Vivipary:

① In several mangrove halophytes, seed germinates while still attached to the parent plant. this phenomenon is called vivipary.

② The young seedling develops before detachment and quickly establishes itself in muddy saline soil after falling.

Xerophyte

Xerophytes are the plants that grow in dry and arid habitats where water availability is very limited. These plants are mainly found in deserts, sandy areas and dry rocky areas. Since excessive water loss is harmful, xerophytes possess several ecological and physiological adaptations to conserve water and survive under ~~draug~~ drought conditions.

① Root adaptation –

① Xerophyte possess a very well-developed root system. In many plants roots grow deep into the soil to absorb underground water.

② Some xerophytes also develop extensive lateral roots that spread widely near the soil surface to absorb rain water quickly.

③ Root hairs are numerous and help in efficient water absorption.

② Stem adaptation –

① The stem of xerophytes are often thick, green and succulent. Succulent stems store water for ~~using~~ use during dry periods.

② In many xerophytes such as cactus, leaves are reduced to spines and stem perform photosynthesis.

③ The stem is covered with thick waxy cuticle which reduces transpiration.

③ Leaf adaptation –

① leaves are usually small, reduced, rolled or modified into spines to minimize surface area and reduce water loss.

② In some plants, leaves become leathery and thick

③ Many xerophytes possess hairy leaves which reduce the effect of hot winds and decrease transpiration.

④ cuticle adaptations-

The epidermis is covered by a very thick cuticle made of wax. This thick cuticle acts as a protective layer and greatly reduces evaporation of water form the plant surface.

⑤ Stomatal Adaptations-

Stomata are few in numbers and usually sunken below the epidermal surface. Sunken stomata reduce direct exposure to dry air and minimize transpiration. In some xerophytes, stomata remain closed during the daytime & open at night.

⑥ Mechanical tissue adaptations-

Mechanical tissues are highly developed in xero-phytes. Thick walled sclerenchyma tissues provide strength and protect the plant from excessive heat and drying conditions.

⑦ Water storage Adaptations-

Many xerophytes develop special water storage tissue in stems, leaves or roots. These tissue store large quantity of water during rainy periods for use in draught conditions.

⑧ Physiological adaptations-

Xerophytes possess physiological mechanism to conserve water. many plants perform photosynthesis with minimum water loss.

Some xerophytes show reduced metabolic activity during extreme draught conditions.

Significance of suncane stomata -

Suncane stomata are stomata located in small ~~pitts~~ pits below the leaf surface. They are commonly found in xerophytic plants adapted to dry environments.

① Reduces transpiration -

> Suncane stomata reduce the rate of water loss by transpiration.

> The pits trap moist air around the stomata, decreasing the evaporation.

② Conserve water -

> It helps xerophytes to survive in arid & desert conditions where water is low.

③ Protection from hot wind -

> The depressed position protects stomata from direct exposure to dry & hot winds.

④ Maintains humidity around stomata -

> Water vapour accumulates inside the stomatal cavity creating a humid microenvironment.

⑤ Prevents excessive water loss during daytime -

> This structure prevents excessive water loss during high temperature & intense sunlight.

⑥ Increase survival in dry Habitats -

> It is an important xerophytic adaptation that improves drought resistance.

Examples -

① Nerium oleander.

② Pinus sylvestris.

Ø Succulent Xerophytes-

Succulent xerophytes are the plants that store large amounts of ~~oxygen~~ H_2O in their stem, leaves or roots.

Characteristics

1) Plant body becomes thick, fleshy and juicy due to water storage.
2) leaves are often reduced into spines to reduce transpiration.
3) Stem usually becomes green & perform photosynthesis
4) Thick cuticle & sunkane stomata ⊛.
5) Extensive root system helps absorb water quickly.

⊛ Stored water helps the plant survive long periods of draughts.

ex- Aloe vera,

Ø Non-Succulent Xerophytes-

Non-succulent xerophytes do not store much water but survive through protective adaptations that reduce water loss.

Characteristics -

1) Plant body is not fleshy.
2) leaves may be small, narrow, rolled, hairy or needle like.
3) Thick cuticle reduce transpiration.
4) ~~Sunca~~ sunken stomata are common.
5) Roots are highly developed & deep.
6) Mechanical tissues are well developed.

⊛ These plant conserve available water efficiently instead of storing it.

ex- Calotropis gigantea.

Feature	Succulent Xerophytes	Non-Succulent xerophytes
1) Water storage	Store large amount of water.	Donot store much water.
2) Plant body	Thick & fleshy	Hard & Non-fleshy.
3) Main adaptation	water storage	water conservation.
4) leaves	often reduced to spines.	small & narrow leaves
5) Stem	usually green & photosynthetic	mostly woody or hard
6) example	Aloevera.	calotropis gigantea.

Ø Aerenchyma

Aerenchyma is a special type of parenchymatous tissue containing large air filled spaces. It is commonly found in hydrophytes or aquatic plants.

Features of aerenchyma

① Cells are loosely arranged with large intercellular air cavities.

② present mainly in stem, roots & leaves of aquatic plants.

③ Tissue is light & spongy in nature.

Functions of aerenchyma-

① Helps in buoyancy and floating of aquatic plant

② Faciliates gaseous exchange b/w submerged plant parts & the atmosphere.

③ Supplies O_2 to submerged tissues.

④ Reduces plant weight & helps survival in water-logged conditions.

example - Eichhornia crassipes.

∅ Pneumatophores–

pneumatophores are specialized negatively geotropic aerial roots found in mangrove plants growing in marshy and waterlogged soils.

Features–

① These roots grow vertically upword from underground roots.
② They project above the soil or water surface.
③ Surface contain small pores called lenticels for gaseous exchange.

Functions–

① Helps in respiration in oxygen-deficient muddy soils.
② Absorb oxygen directly from the atmosphere.
③ Facilitate gaseous exchange through lenticels.
④ Help mangrove plants survive in swampy habitats

example– Sonneratia alba.

∅ Vivipary –

Vivivary is a special type of seed germination in which the seed begins to germinate while still attached to the parent plant.

It is commonly seen in mangrove plants growing in saline and marshy habitats.

Features–

1) Seed germinate inside the fruit before detachment from the parent plant.
2) The radicle elongates & forms a seedlings while attached to the plant.
3) Helps the seedlings to establish quickly in muddy soil after falling.

Importance

① prevents seed death in saline & waterlogged conditions
② Ensure rapid establishment of seedlings.
③ Reduces chances of dispersal by tides & water currents.
④ Helps mangrove plants survive in unstable muddy habitats.

example –

Rhizophora mucronata.

Ø water Conservation mechanisms of Xerophytes–

Water conservation mechanisms of xerophytes are special structural, anatomical & physiological adaptations that reduce water loss & help plants survive in dry & arid environments.

① Thick waxy cuticle reduces transpiration.
② Sunken stomata decrease water loss by trapping moist air.
③ Reduced leaf size or leaf modified into spines minimize transpiration.
④ Hairy leaf surface reduces evaporation & protects from hot winds.
⑤ Rolled or folded leaf reduce exposed surface area.
⑥ Succulent stems or leafs store water for future
⑦ Excessive root system absorb maximum available water.
⑧ Stomata remain closed during the hottest part of the day in many xerophytes.

examples –

Nerium oleander

∅ Salt tolerance mechanisms of Halophytes-

Halophytes are plants adapted to grow in saline soils or salty water conditions. They possess special mechanisms to tolerate excess salt.

① Salt excretion-

> Some halophytes remove excess salt through special salt glands or salt hairs present on leaves.

example- Avicennia marina.

② Salt Accumulation-

> Excess salts are stored in vacuoles of cells to prevent damage to the cytoplasm.

③ Succulence-

> Stems & leaves become thick & fleshy due to water storage.

> Stored water dilute salt concentration inside tissues.

④ Selective absorption of ions-

> Roots selectivey absorb useful minerals while restricting toxic salts like sodium chloride (NaCl)

⑤ Osmotic Adjustment-

> Cells maintain high osmotic concentration by accumulating organic substances such as sugers & amino acids.

> Help absorb water from saline soil.

⑥ Shedding of salt rich leaves-

> old leaves contain excessive salt shed from the plant body.

⑦ Vivirary-

Seed germinate while attached to the parent plant increase survival in saline habitats.

⑧ Pneumatophores-

Specialized respiratory roots helps in gaseous exchange in saline waterlogged soils.

	Hydrophyte	Halophyte	Xerophyte
Roots	① Roots poorly developed ② Root hair absent/few ③ mainly Anchorage rather than absorption ④ Free floating highly reduced.	① Root well developed ② Pneumatophore ③ Small pore lenticel	① Root well developed ② Deep root ③ lateral root ④ Abundant root hair
Stem	① Soft, flexible, spongy ② Aerenchyma ③ Mechanical tissue ④ flexible → water current	① thick, succulent ② succulent → water ↓ salt conch ↓	① thick, succulent ② water store ③ leaves → spine ④ stem → waxy cuticle
leaf	① waxy leaf → wetting & decay ② Broad, Flat → floating ③ thin, narrow → submerged ④ Long petiole	① thick, succulent ② Succulent → salt conc^{n} ↓ ③ Reduce leaf area	① thick, leathery ② succulent ③ small, reduce, rolled
Stomata	① floating → ⊕ only in upper surface ② Submerged → ⊖	① Few, sunken	① Few, sunken
Cuticle	① Very thin ② often ⊖	① Thick ⑪ waxy covering	① Thick ⑪ waxy covering
MT	① Scheleren chyma reduced	⑩ Scheleren chyma highly developed	① Highly developed
VT	⑪ poorly developed	⑪ Highly developed	⑪ Highly developed
Special feature		① vivipary ⑪ salt secreting gland	① water storage adaptation.

Ø Physiological adaptations in hydrophytes -

Ⓐ Reduced transpiration -

① Since water is abundant, hydrophytes do not need to conserve it. aquatic habitat has high humidity → water vapour gradient low ↓ low transpirⁿ

② Cuticle is thin or absent, & stomata remain open for longer periods.

Ⓑ Efficient gaseous exchange -

① Large air spaces (aerenchyma) helps to transfer port oxygen to submerged plants.

② Dissolved gases are absorbed directly from water.

Ⓒ Low mechanical & vascular tissue development -

① Water support the plant body so mechanical tissue like scheleranchyma is poorly developed.

② Xylem (vascular tissue) is poorly developed as water is readily available & therefore the ~~conduction of~~ requirement of conduction of water is very low.

Ⓓ Anaerobic respiration tolerance -

① Roots in muddy water often face oxygen deficiency.

② Cells can tolerate low oxygen conditions & perform limited anaerobic ~~condition~~ respiration.

Ⓔ Osmotic ~~Ad~~justment -

① Cell sap concentration is usually low because water absorption occurs easily.

② Minimal energy is required for water uptake.

Ⓕ Photosynthetic adaptation -

① Submerged leaf ~~can do~~ perform photosynthesis using dissolved carbondioxide & bicarbonates.

② Chloroplasts are often concentrated near the upper surface for efficient light absorption.

Ⓖ Nutrient absorption through body surface -

① Many hydrophyte absorb minerals not only through roots but also through leaf & stems directly from water.

∅ Anatomical Adaptations of Hydrophytes-

① Well developed aerenchyma-

(i) Large intercellular air spaces are present in roots, stems & leaves

(ii) Help in buoyancy & gaseous exchange.

② Thin epidermis-

(i) Epidermal cells are usually thin-walled.

(ii) Cuticle is very thin or absent.

③ Reduced vascular tissue-

(i) Xylem is poorly developed because water is readily available.

(ii) Phloem remains comparatively better developed.

④ Poorly developed mechanical tissue-

(i) Sclerenchyma and collenchyma are reduced or absent.

(ii) Water provides support to the plant body.

⑤ Large cortex-

(i) Cortex is broad and contains numerous air chambers.

⑥ Stomatal arrangement-

(i) Floating leaves have stomata only on the upper epidermis

(ii) Submerged ~~leaf~~ leaves ~~are~~ generally lack stomata.

⑦ Mesophyll differentiation-

(i) In floating leaves, mesophyll is differentiated into palisade & spongy tissues.

(ii) In submerged leaves, mesophyll is often undifferenti-ated.

⑧ ~~Stomatal arrangement~~ - Chloroplast distribution-

(i) Chloroplasts are abundant in photosynthetic tissues and often concentrated near the surface.

ø Physiological Adaptations of Halophytes-

① Osmotic adjustment -

ⓘ Cells accumulates compatible solutes (osmoprotectants) such as proline, glycine betaine, sugers & organic acids.

ⓘⓘ This lowers the osmotic potential of the cell sap & enables water absorption from saline soils.

② Salt regulation -

ⓘ Excess salts are either excluded by roots or secreted through salt glands and salt hairs present on leaves.

ⓘⓘ This prevents toxic accumulation of salts in the cytoplasm.

③ Ion ~~Compartmentization~~ - Compartmentalization -

ⓘ Na^+ & Cl^- ions are stored in vacuoles, while cytoplasm maintains a relatively normal ionic balance.

④ Succulence -

ⓘ Many halophytes store large amounts of water in their tissues, which helps dilute the salt concentration within cells.

⑤ Stomatal regulation

ⓘ Stomata regulate transpiration efficiently to reduce excessive water loss under physiological draught conditions.

⑥ Tolerance to physiological drought -

ⓘ Although water is present in the soil, its high salt concentration makes it difficult to absorb.

ⓘⓘ Halophyte can continue normal metabolic activity under such conditions.

⑦ Enzyme & metabolic adaptation -

ⓘ Their enzyme & protein remain functional even at high salt concentrations, allowing photosynthesis & respiration to continue.

Ø Anatomical adaptations of Halophytes-

① Thick cuticle-

The epidermis is covered with a thick cuticle to reduce water loss.

② Multiple epidermis or hypodermis-

Many halophytes possess multilayered epidermis or a well-developed hypodermis for protection.

③ Succulent tissue-

Large thin walled parenchymatous cells store water & help dilute excess salts.

④ Well developed water storage tissue-

Specialized parenchyma tissues are present for water storage.

⑤ Salt glands & salt hairs-

Specialized structures help in the secretion of excess salts from the plant body.

⑥ Compact mesophyll-

Mesophyll tissue have fewer intercellular spaces, minimizing water loss.

⑦ Well developed vascular tissues-

Especially the xylem is well developed to fascilitate efficient water transport.

⑧ Strong mechanical tissue-

Collenchyma & sclerenchyma are well developed providing support & rigidity.

∅ Physiological Adaptations of Xerophytes-

① **High osmotic concentration-**

Cell sap has a high concentration of solutes, enabling efficient absorption of water from dry soils.

② **Reduced transpiration-**

Stomata remain closed during the hottest part of day to minimize water loss.

③ **CAM Photosynthesis-**

Many ~~stomata~~ xerophytes stomata open at night & fix carbon dioxides ~~in~~ into organic acids

During day, stomata remain closed & the stored CO_2 is used for photosynthesis, reducing water loss.

④ **Water-storage capacity-**

Cells maintain high water-holding capacity & use stored water during ~~s~~ draught conditions.

⑤ **Draught tolerance-**

protoplasm can withstand dehydration & resume normal activity when water becomes available.

⑥ **Efficient stomatal regulation-**

Guard cells repond ~~rea~~ rapidly to environmental changes, controlling the opening & closing ~~to~~ of stomata

⑦ **Accumulations of osmoprotectants-**

Compounds such as proline, sugers & amino acids accumulates in cells, helping maintain osmotic balance & protecting cellular structures.

⑧ **Low rate of respiration & metabolism during stress-**

Metabolic activity may decrease during severe draught, conserving energy & water.

∅ Anatomical Adaptations of Xerophytes-

① **Thick cuticle-**

The epidermis is covered by a thick layer/ waxy cuticle to minimize transpiration.

② **Multiple epidermis-**

Many xerophyte possess a multilayered epidermis that provides additional protection against water loss.

③ **Sunken stomata-**

Stomata are located in depressions (stomatal crypts) reducing the rate of transpiration.

④ **Well developed vascular tissue-**

Xylem & phloem are highly developed to ensure efficient transport of water & food.

⑤ **Strong mechanical tissue-**

Collenchyma & sclerenchyma are well developed giving rigidity & protection.

⑥ **Water storage tissue-**

Specialized parenchymatous tissue to store water, especially in succulent xerophytes.

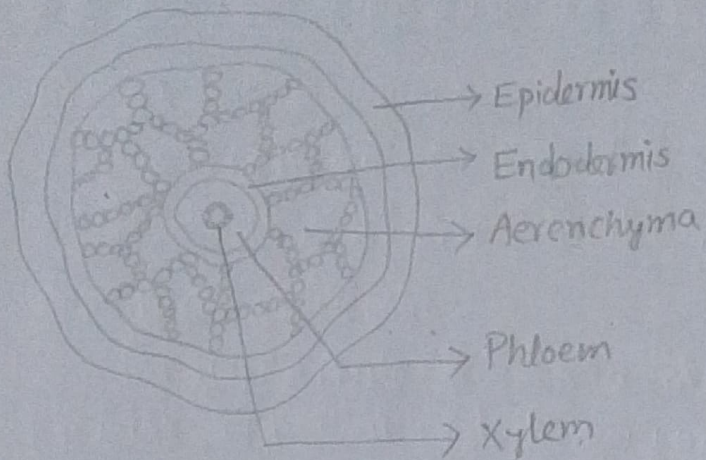


Fig: TS of Hydrilla stem. (showing Aerenchyma)

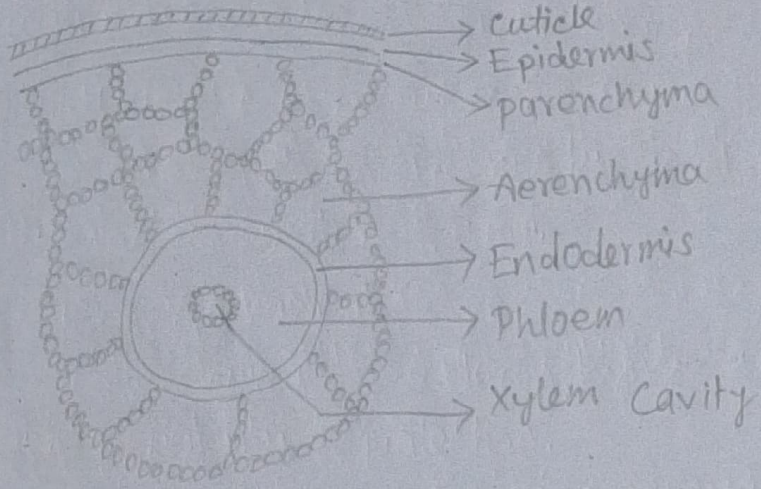


Fig: TS of Hydrila stem, including thin cuticle
(it shows thin cuticle)

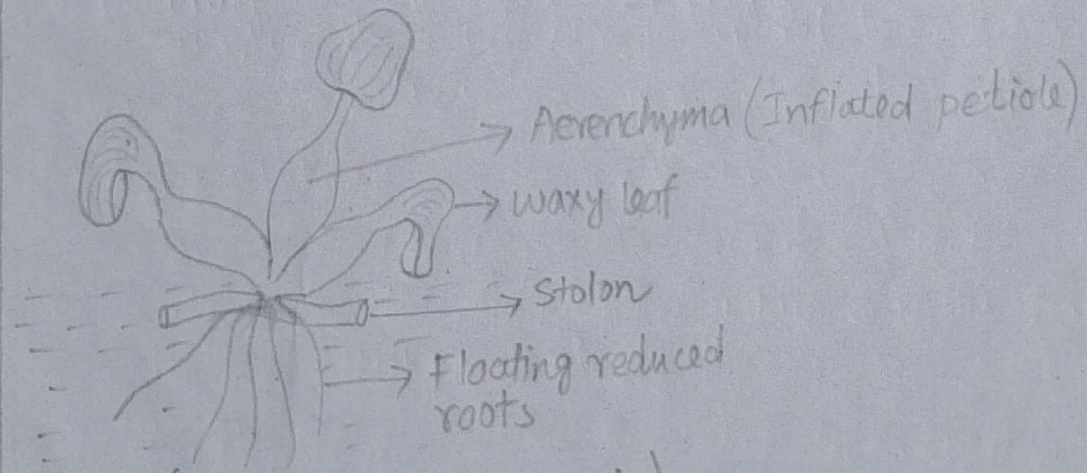


Fig: ✗Eichhornia (Hydrophyte)
free floating

✓Free floatig Hydrophyte (Eichhornia)

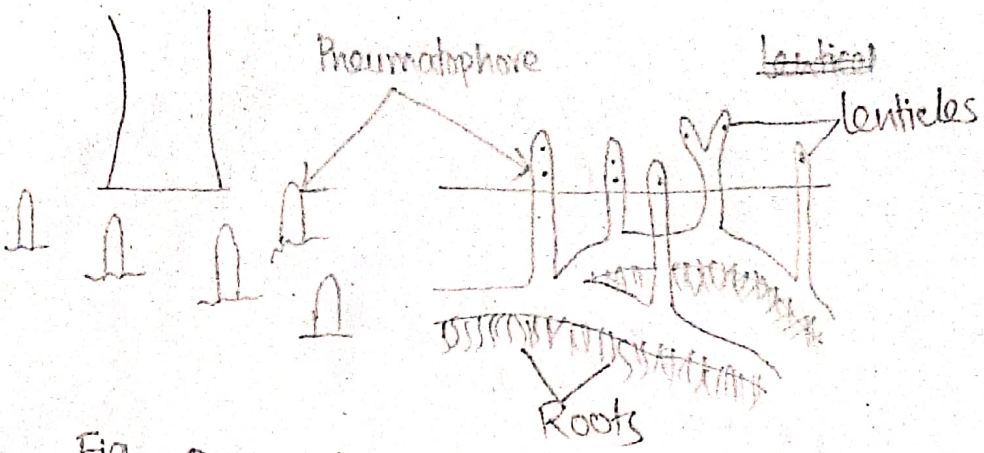


Fig- Pneumatophores of mangrove plants.

